

Euler's Totient Function and Euler's Theorem

1 What $\phi(n)$ Counts

Definition

For $n > 1$, Euler's totient function $\phi(n)$ counts how many integers from 1 to $n - 1$ are relatively prime to n :

$$\phi(n) = |\{r : 1 \leq r < n, \gcd(r, n) = 1\}|.$$

The numbers counted by $\phi(n)$ are the invertible residues modulo n . That means each of them has a multiplicative inverse modulo n .

Why This Matters

Euler's theorem says that if a is invertible modulo n , then a certain power of a becomes 1 modulo n . This is useful for reducing large powers.

2 First Example

Find $\phi(12)$

List the numbers from 1 to 11, then keep only the ones relatively prime to 12.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11.

The numbers relatively prime to 12 are

1, 5, 7, 11.

So

$$\phi(12) = 4.$$

Number	Reason
1	$\gcd(1, 12) = 1$, so count it.
5	$\gcd(5, 12) = 1$, so count it.
7	$\gcd(7, 12) = 1$, so count it.
11	$\gcd(11, 12) = 1$, so count it.

3 Formula for $\phi(n)$

Totient Product Formula

If the distinct prime divisors of n are $p_1, p_2, \dots, p_k$, then

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_k}\right).$$

Equivalently,

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Common Mistake

Use only the distinct prime divisors of n .

For example, if

$$n = 2^3 \cdot 3^2,$$

then the formula uses only 2 and 3, not repeated copies of 2 and 3.

4 How to Compute $\phi(n)$

Totient Calculation Steps

1. Factor n into primes.
2. Keep only the distinct prime divisors.
3. Start with n .
4. For each distinct prime p dividing n , multiply by $(1 - 1/p)$.
5. Simplify.

Example: $\phi(45)$

Factor:

$$45 = 3^2 \cdot 5.$$

The distinct prime divisors are 3 and 5. Therefore

$$\phi(45) = 45 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right).$$

Simplify:

$$\phi(45) = 45 \cdot \frac{2}{3} \cdot \frac{4}{5}.$$

$$\phi(45) = 24.$$

Answer

$$\phi(45) = 24.$$

More Examples

n	Factorization	$\phi(n)$
7	7	$\phi(7) = 7(1 - \frac{1}{7}) = 6$
9	3^2	$\phi(9) = 9(1 - \frac{1}{3}) = 6$
12	$2^2 \cdot 3$	$\phi(12) = 12(1 - \frac{1}{2})(1 - \frac{1}{3}) = 4$
20	$2^2 \cdot 5$	$\phi(20) = 20(1 - \frac{1}{2})(1 - \frac{1}{5}) = 8$

5 Prime and Prime-Power Cases

Useful Special Cases

If p is prime, then

$$\phi(p) = p - 1.$$

If p is prime and $e \geq 1$, then

$$\phi(p^e) = p^e - p^{e-1} = p^{e-1}(p - 1).$$

Why does the prime-power formula work? Among

$$1, 2, \dots, p^e - 1,$$

the only numbers not relatively prime to p^e are multiples of p . There are p^{e-1} such multiples below p^e . So

$$\phi(p^e) = p^e - p^{e-1}.$$

6 Euler's Theorem

Euler's Theorem

If

$$\gcd(a, n) = 1,$$

then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

The condition $\gcd(a, n) = 1$ is required. Without it, a may not be invertible modulo n , and the theorem may fail.

7 Proof of Euler's Theorem

Short Proof

List all $\phi(n)$ invertible residues modulo n , and multiply each of them by a .

Since $\gcd(a, n) = 1$, multiplication by a only rearranges the invertible residues. It cannot merge two different residues, because

$$ax \equiv ay \pmod{n} \Rightarrow x \equiv y \pmod{n}.$$

So the product after multiplication is congruent to the original product:

$$a^{\phi(n)} \cdot (\text{original product}) \equiv \text{original product} \pmod{n}.$$

The original product is invertible modulo n , so we cancel it:

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

8 Fermat's Little Theorem

Prime Modulus Version

If p is prime and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

This is Euler's theorem with $n = p$, because

$$\phi(p) = p - 1.$$

9 Using Euler's Theorem

Problem

Find the remainder when

$$7^{50}$$

is divided by 20.

First check:

$$\gcd(7, 20) = 1.$$

So Euler's theorem applies.

Compute

$$\phi(20) = 20 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 8.$$

Thus

$$7^8 \equiv 1 \pmod{20}.$$

Reduce the exponent:

$$50 = 8(6) + 2.$$

So

$$7^{50} = 7^{8 \cdot 6 + 2} = (7^8)^6 \cdot 7^2.$$

Modulo 20:

$$7^{50} \equiv 1^6 \cdot 49 \equiv 9 \pmod{20}.$$

Answer

$$7^{50} \equiv 9 \pmod{20}.$$

10 Another Quick Example

Problem

Evaluate

$$5^4 \pmod{12}.$$

Since

$$\gcd(5, 12) = 1$$

and

$$\phi(12) = 4,$$

Euler's theorem gives

$$5^4 \equiv 1 \pmod{12}.$$

Answer

$$5^4 \equiv 1 \pmod{12}.$$

11 Final Review

Core Facts

$$\begin{aligned} \phi(n) &\text{ counts residues } 1 \leq r < n \text{ with } \gcd(r, n) = 1. \\ \phi(n) &= n \prod_{p|n} \left(1 - \frac{1}{p}\right). \\ \phi(p) &= p - 1 \text{ for prime } p. \\ \phi(p^e) &= p^{e-1}(p - 1). \\ \gcd(a, n) = 1 &\Rightarrow a^{\phi(n)} \equiv 1 \pmod{n}. \end{aligned}$$